

LEE HEALTH POLICY & PROCEDURES

CENTRAL VENOUS ACCESS DEVICE CATHETERS - ADULT			LOCATOR NUMBER																		
T Y P E	☐ System-wide - A formal statement of values, intents (policy), and expectations (procedure) that applies to every employee throughout the System. ☒ Multidisciplinary/Interdisciplinary - A formal statement of values, intents (policy), and expectations (procedure) that applies to more than one discipline and is usually of a clinical nature. Check below all areas to which this applies. ☐ Departmental - A formal statement of values, intents (policy), and expectations (procedure) exclusive to a particular department or group of people within a department at one or multiple locations that does not impact any other area.	CHAPTER: M03 TAB: 10 POLICY #: 112																			
Disciplines / locations to which this interdisciplinary policy applies:																					
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As Needed:																					
			Date:																		

PURPOSE:

To provide guidelines for the insertion, care, maintenance, and removal of central venous access devices (CVADs) in the adult population.

For care, maintenance, and removal of the following vascular access devices, please refer to the appropriate policies:

[M03 10 357](#) Hemodialysis Catheters Used for CRRT, Apheresis, or Dialysis

[M03 10 427](#) Implanted Central Venous Access Device- Adult Care and Maintenance

[M03 10 448](#) IV Therapy- General Guidelines

[M18 00 048](#) Central Venous Access Device Care and Maintenance – Pediatric / NICU

POLICY:

- A. Only licensed practitioners with delineated privileges and demonstrated competencies may insert central venous access devices.
- B. Licensed nursing staff will complete initial education related to the care and maintenance of CVADs.
- C. Bundles shall guide the standards and performance in insertion, care, and maintenance for Central Venous Access Devices (CVADS). Refer to Appendix A-1 for Central Venous Access Device (CVAD) Bundle Elements and Appendix A-2 for CLABSI Maintenance Bundle Care Descriptions, Form #0544.
- D. Remove CVAD within 24-48 hours, or when medically feasible, when the catheter is inserted under suboptimal aseptic conditions (i.e. emergent situations).
- E. Patients presenting to the to the health system with a present on admission CVAD shall have a consult to vascular access team. The vascular access team will assess the site and confirm placement verification for the CVAD to be used.
 - 1. Patients with tunneled CVADs do not require tip placement verification unless signs of malposition are present.
 - 2. For PICC line clearance refer to policy M03 10 711 Peripherally Inserted Central Catheter (PICC), Ultrasound Guided Insertion– Adult.
 - 3. EXCLUSION: Outpatients being cared for in the inpatient setting - Patency verification will be confirmed with blood return and assessment of function.

CXR confirmation is also not needed when the patient is traveling from inpatient unit to inpatient unit within the Lee Health System.
- F. All patients entering the healthcare system with a CVAD should be assessed for signs and/or symptoms of infection. When signs and/or symptoms of infection are present, notify provider.
 - 1. Signs and symptoms of CVAD infections include erythema, edema, pain, tenderness or drainage, fluid in the subcutaneous pocket, and/or tunnel, induration at the exit site or over the pocket, drainage, or skin breakdown at the insertion site, and/or body temperature elevation.
 - 2. If clinically warranted, obtain peripheral blood cultures (physician or provider order required).
 - 3. Blood cultures should not be obtained from CVADs as not considered best practice.
 - a) If blood cultures are ordered to be drawn from a CVAD, only nurses from ICU, ED, oncology, or VAN team are permitted to obtain them from the line.
- G. Blood draws from CVADs require a physician or provider order. Blood draws from any CVAD require a second nurse to witness the draw. Documentation of second nurse witness in the

electronic medical record is required. Hospital at Home will not require a second nurse witness from the home.

- H. TPN requires a dedicated port which will not be manipulated or accessed.
- I. The distal port of a multi-lumen CVC shall be used for blood transfusions, colloid fluid, high volume fluid administration. The distal port of a multi-lumen CVC is the recommended lumen for CVP monitoring.
- J. Limit the use of add-on devices whenever possible to decrease excessive manipulations, accidental disconnections or misconnections, and risk of contamination.
 - 1. Stopcocks use is only when necessary to zero a CVAD or infuse fluids. Use a stopcock with an integrated needleless connector rather than a solid cap or replace the solid cap with a needleless connector to maintain a closed system whenever possible.
- K. All patients with a CVAD shall have daily chlorhexidine gluconate treatment. Refer to policy M03 01 115 Chlorhexidine Gluconate (CHG) Treatment – Adult.
- L. Patient shall not be discharged with the following central venous access devices without a physician order:
 - 1. Temporary dialysis catheter (dual lumen or dual lumen with pigtail)
 - 2. Non-tunneled central venous catheter
 - 3. Accessed IVAD (mediport)
- M. Patients may be discharged with the following long-term central venous access devices:
 - 1. PICC when ongoing treatment plan in place
 - 2. Tunneled dialysis catheter
 - 3. Powerline (tunneled central venous catheter)
- N. Site Assessment, Care, and Maintenance
 - 1. When accessing a CVAD, the clinician is to perform hand hygiene, don clean gloves, and scrub the hub with an alcohol pad for 15 seconds prior to each entry of the line.
 - 2. All CVADs shall have a luer-lock single needleless connector in place. CVAD lumens not in use shall be clamped when a clamp is present. Single needleless connectors should be changed every seven days, if residual blood or debris within the needleless connector, and/or if connector is compromised.
 - 3. All CVADs shall have an antimicrobial patch placed at insertion site unless there is a documented sensitivity. Antimicrobial patches should have 360° contact with skin around catheter.
 - a) Upon insertion of catheters and when bleeding occurs, StatSeal topical hemostatic discs and powder may be used.

4. The RN will assess the CVAD every 4 hours to ensure dressing is occlusive and the patient has no signs or symptoms of malposition, infection, or complications and document CVAD assessment in the electronic health record. Hospital at Home RN will assess the CVAD every four hours between the hours of 0700 and 2200 virtually and/or in-person during the Field RN visit.
5. When initiating the use of a new CVAD, new tubing and fluid sets shall be used. For further tubing management and medication labeling refer to policy M03 01 880 Tubing Connections (Luer) and [M03 03 247 Expiration Dating for Medications and Solutions](#).
6. Flushing Guidelines
 - a) CVAD should be assessed for patency (flushed and aspirated for blood return) prior to each infusion to assess catheter function and prevent complications.
 - b) CVADs shall be flushed with prefilled flush syringes of 0.9% normal saline 10 ml size.
 - c) Use new, unopened, single-use 0.9% saline syringes for flushing before and after medications. Do not use same syringe on more than one catheter lumen entry.
 - d) Medication shall not be diluted in a saline syringe.
 - e) Use of push-pause technique for flushing is vital in maintaining patency and preventing catheter occlusions.
 - f) Do not forcibly flush any CVAD. If resistance is met and/or no blood return noted, take further action including notifying provider and/or consulting the VAN team.
 - g) Flush the CVAD with 10ml 0.9% normal saline following the administration of IV push medications.
 - h) Use positive pressure techniques to minimize blood reflux into the CVAD lumen.
 - i) Leave a small amount (0.5 to 1.0 ml) of saline solution in the syringe to avoid compression of the plunger rod to prevent blood reflux.
 - j) Disconnect syringe or tubing from positive displacement needless connector before clamping the line to allow positive displacement to occur.

O. CVAD Dressings & Securement

1. CVAD dressing changes may only be performed by those with documented competencies.
2. CVAD dressing change kits are preferred.
3. Transparent dressings are changed every seven days and anytime a dressing's integrity is compromised. Examples of compromised dressing include:
 - a) Dressing is nonocclusive.

- b) Edges appear to be lifted or detached.
 - c) The transparent portion of the dressing is visibly soiled (presence of moisture, drainage, or blood)
- 4. A sterile gauze dressing is only to be used when a patient is allergic to a transparent dressing. A gauze dressing shall be changed every 48 hours.
 - 5. Sterile alcohol-based chlorhexidine solution is the preferred skin antiseptic agent for CVAD site disinfection. If there is a contraindication to chlorhexidine solution, povidone-iodine or 70% alcohol may be used.
 - 6. CVADs are secured to prevent complications including dislodgement and motion at the insertion site. The preferred securement method is a suture free securement device ensuring the exit site has room for antimicrobial patch application. Sutures may also be used but are associated with higher complication rates such as needlestick injury, support the growth of biofilm, and increase the risk of catheter-associate bloodstream infection.
 - 7. Dressings must be labeled appropriately with date, time, and initials.
 - 8. Dressing changes to be documented in the electronic health record.

PROCEDURE:

A. Insertion of CVADs

- 1. CHG treatment shall be done prior to line insertions (exclusion: emergent line placement).
- 2. A central line insertion checklist shall be used for all insertions.
- 3. Maximal sterile barriers and aseptic technique, including sterile gown, sterile gloves, and a large sterile drape are used for insertion shall be used.

B. Removal of CVADs

- 1. A physician or provider order is required for removal of a CVAD.
- 2. A tunneled CVAD is removed ONLY by a licensed independent practitioner (LIP).
- 3. A non-tunneled CVAD may be removed by a registered nurse or interventional radiology technician who has received training and has documented competency. Refer to [Job Instruction: Adult Non-Tunneled Central Venous Access Device Removal](#) and [FM# 1205](#) for device removal steps.

a. For PICC line removal refer to Peripherally Inserted Central Catheter (PICC) Ultrasound Guided Insertion – Adult policy M03 10 711 and [FM#1204](#) for device removal steps.

C. Procedure Index:

Appendix A-1: Central Venous Access Device Bundle Elements

Appendix A-2: [CLABSI Maintenance Bundle Care Descriptions](#)

Appendix B- [Job Instruction: Adult Non-Tunneled Central Venous Access Device Removal](#)

Appendix C: CVAD Dressing Change

Appendix D: CVAD Dressing Change for Bleeding Site

Appendix E: Adult Blood Sampling from a CVC, Vacutainer Method

Appendix F: Adult Blood Sampling from a CVC, Syringe Method

Appendix G: Adult Blood Sampling from a CVC In-line Blood Sampling System

RELATED POLICIES:

M03 01 115 Chlorhexidine Gluconate (CHG) Treatment - Adult

M03 01 880 Tubing Connections (Luer)

M03 03 560 Medication Orders, Administration, Documentation and Time Critical Medications

M03 03 700 Parenteral Nutrition Guidelines – Adult

M03 05 074 Blood Collection

M03 05 076 Blood Culture Collection

M03 05 132 Calibration of Pressure Monitoring Lines

M03 10 028 Alteplase (t-PA) Administration for Catheter Occlusions – Adult

M03 10 299 Extravasation-Infiltration of IV Drugs – Intervention and Management

M03 10 357 Hemodialysis Catheters Used for CRRT, Apheresis or Dialysis - Adult

M03 10 427 Implanted Central Venous Access Device for Adults, Care and Maintenance of

M03 10 448 IV Therapy – General Guidelines

M03 10 711 Peripherally Inserted Central Catheter (PICC) – Ultrasound Guided Insertion - Adult

M03 10 605 Midline Catheter - Adult

S11 00 356 Hand Hygiene

M18 00 048 Central Venous Access Devices (CVAD) Care and Maintenance

S25 00 705 Patient Identification / Red Rule

REFERENCES:

Infusion therapy: Standards of practice (9th ed.). (2024). Infusion Nurses Society.

Policies and Procedures for Infusion Therapy: Acute Care (7th ed.) (2024) Infusion Nurses Society.

Appendix A-1

Central Venous Access Device Bundle Elements

1. Hand hygiene

When accessing a CVAD, the clinician is to perform hand hygiene, don clean gloves, and scrub the hub with an alcohol pad for 15 seconds prior to each entry of the line. Refer to policy M11 00 356 Hand Hygiene.

2. Daily chlorhexidine gluconate treatment.

3. CVAD insertion checklist should be used for all insertions.

4. Maximal sterile barrier precautions upon insertion.

Maximal sterile barriers and sterile technique, including sterile gown, sterile gloves, hat, mask, and a large sterile drape are used for insertion.

5. Chlorhexidine skin antisepsis is the preferred

Sterile alcohol-based chlorhexidine solution is the preferred skin antiseptic agent for CVAD site disinfection.

6. Optimal catheter site selection

Catheters with the minimal number of ports or lumens needed for that patient should be selected. Subclavian and internal jugular insertion is preferred.

7. Daily review of CVAD necessity

Evaluate criteria for CVAD necessity and continued need for central access daily.

8. Avoid blood draws from CVAD.

No blood draws from a CVAD unless ordered by a physician. Use blood in lab whenever possible or future draw. Second RN witness of blood draw required.

9. Sutureless securement devices are preferred.

Appendix C

CVAD Dressing Change

Pre-Procedure Preparation

1. Perform hand hygiene before direct contact with patient and subsequently as required throughout procedure steps.
2. Verify patient with two identifiers.
3. Gather supplies, disinfect work area (as applicable).
4. All-inclusive CVAD dressing change kit required.
5. Explain procedure to patient.
6. Assemble supplies on a sterile field.

Procedure

1. Place patient in supine position unless contraindicated so that the insertion site is below the level of the heart.
2. Turn patient's head to opposite side.
3. Perform hand hygiene and don mask.
4. Open central line dressing kit without contamination and assemble supplies on sterile field.
5. Don clean gloves.
6. Remove dressing and antimicrobial patch.
 - a. Consider use of adhesive remover for removing dressing.
 - b. Loosen toward insertion site ensuring catheter stability.
 - c. Anchor catheter and remove securement device (if applicable) avoiding migration.
 - d. Beginning at the device hub and gently pulling the dressing perpendicular to the skin towards insertion site
7. If unable to easily remove dressing and antimicrobial patch:
 - a. Stop and remove clean gloves and perform hand hygiene.
 - b. Don sterile gloves.
 - c. Using dominant hand, secure catheter at insertion site and complete dressing removal along with antimicrobial patch and securement device (if present).
8. Assess insertion site for signs and symptoms of infection as well as external catheter length. If signs and symptoms of infection or catheter dislodgement are present, notify provider.

9. Remove gloves and perform hand hygiene.
10. Don sterile gloves.
11. Cleanse insertion site with sterile chlorhexidine using back and forth motion for 30 seconds and allow to dry.
 - a. If there is a contraindication to chlorhexidine solution, povidone-iodine or 70% alcohol may be used.
12. Apply antimicrobial patch to insertion site ensuring it is around the insertion site circumferentially.
13. Apply skin prep to outer perimeter where dressing will be, do not place over insertion site. Allow skin prep to dry.
14. Apply securement device (if indicated).
15. Apply appropriate sterile dressing over insertion site.
16. For seven-day dressing change, remove and replace needleless connectors.
 - a. Ensure catheter lumens are clamped and remove needleless connector.
 - b. Clean ends of catheter lumens with alcohol using friction for 15 seconds and allow to dry.
 - c. Attach newly primed needleless connectors, assess each lumen for patency with blood return and flush using the push pause method. Clamp lumens if not in use.
 - d. Remove gloves and perform hand hygiene.
17. Label dressing with date, time, and initials.
18. Remove gloves, perform hand hygiene.
19. Document procedure in electronic health record.

Appendix D

CVAD Dressing Change for Bleeding Site

Pre-Procedure Preparation

1. Perform hand hygiene before direct contact with patient and subsequently as required throughout procedure steps.
2. Verify patient with two identifiers.
3. Gather supplies, disinfect work area (as applicable).
 - a. All-inclusive CVAD dressing change kit required.
 - b. Hemostatic powder or disc
 - c. Sterile 4x4
4. Assemble supplies on a sterile field.
5. Explain procedure to patient.

Procedure

1. Turn patient's head to opposite side.
2. Perform hand hygiene and don mask.
3. Don clean gloves.
4. Remove dressing and antimicrobial patch/ hemostatic agent.
 - a. Consider use of adhesive remover for removing dressing.
 - b. Loosen toward insertion site ensuring catheter stability.
 - c. Anchor catheter and remove securement device (if applicable) avoiding migration.
 - d. Beginning at the device hub and gently pulling the dressing perpendicular to the skin towards insertion site
5. If unable to easily remove dressing and antimicrobial patch:
 - a. Stop and remove clean gloves and perform hand hygiene.
 - b. Don sterile gloves.
 - c. Using dominant hand, secure catheter at insertion site and complete dressing removal along with antimicrobial patch and securement device (if present).
7. Wipe blood from exit site using sterile 4x4 gauze.
8. Inspect catheter for any kinks, leaks, or breakage.

9. Assess insertion site for signs and symptoms of infection as well as external catheter length. If signs and symptoms of infection or catheter dislodgement are present, notify provider.
10. Observe for oozing/bleeding at insertion site. If present:
 - a. Place head of bed at 45 degrees if condition allows.
11. Remove gloves and perform hand hygiene.
12. Don sterile gloves.
13. Cleanse insertion site with sterile chlorhexidine using back and forth motion for 30 seconds and allow to dry.
 - a. If there is a contraindication to chlorhexidine solution, povidone-iodine or 70% alcohol may be used.
14. If using StatSeal hemostatic disc:
 - a. Place disc with label side up around catheter. The disc slit should be perpendicular to catheter and slit edges should approximate each other to the greatest extent possible.
 - b. Maintain continuous, semi-occlusive pressure over disc for 2 minutes. Extend pressure times as clinically necessary.
15. If using StatSeal hemostatic powder:
 - a. Remove paper backing from powder containment device to expose adhesive. Center powder containment device around site with adhesive side down.
 - b. Gently shake tube prior to opening. Invert tube to pour powder; do not squeeze tube.
 - c. Fill PCD to brim with powder; do not overfill. Ensure powder completely covers and surrounds site. Place PCD foam insert over powder to cover.
 - d. Maintain continuous, semi-occlusive pressure over site for 2 minutes. Extend pressure times as clinically necessary.
16. If oozing/bleeding continues, secure site with gauze covered by a transparent dressing and notify.
 - a. Interventional Radiology (IR) for orders if catheter was inserted by IR.
 - b. Inserting physician if inserted in OR.
 - c. Intensivist if inserted in the ICU.
17. If hemostasis achieved, apply skin prep to outer perimeter where dressing will be, do not place over insertion site. Allow skin prep to dry.
18. Apply securement device (if indicated).
19. Apply appropriate sterile dressing over insertion site.

20. For seven-day dressing change, remove and replace needleless connectors.
 - a. Ensure catheter lumens are clamped and remove needleless connector.
 - b. Clean ends of catheter lumens with alcohol using friction for 15 seconds and allow to dry.
 - c. Attach newly primed needleless connectors, assess each lumen for patency with blood return and flush using the push pause method. Clamp lumens if not in use.
 - d. Remove gloves and perform hand hygiene.
21. Primary nurse to reassess site hourly X 4 for bleeding. If bleeding returns, contact same physician as above and VAN. Repeat this protocol.
22. StatSeal hemostatic disc and powder may remain in place until next dressing change/up to 7 days.
23. Label dressing with date, time, and initials.
24. Remove gloves, perform hand hygiene.
25. Document procedure in electronic health record.

Appendix E

Adult Blood Sampling from a CVAD, Vacutainer Method (NOT recommended for PICC lines)

No blood draws from a central line (unless ordered by physician or provider); use blood in lab whenever possible or future draw. Refer to Blood Collection policy for details on order of draws.

Pre-Procedure Preparation

1. Perform hand hygiene before direct contact with patient and subsequently as required throughout procedure steps.
2. Verify patient with two identifiers.
3. Gather supplies, disinfect work area (as applicable).
4. Explain procedure to patient.
5. Ensure second RN is present. Hospital at Home will not require second RN presence.
6. Stop all infusions through the catheter, clamp lumens, and/or stop infusions as appropriate.

Procedure

1. Scrub the hub with an alcohol pad for 15 seconds prior to each entry of the line and allow to dry.
2. Flush lumen with 0.9% 10 ml sodium chloride before obtaining blood sample using push pause method.
3. Withdraw discard of 5 ml into syringe. If collecting coagulation labs through a catheter with heparin, withdraw discard of 10 ml. Do not reinfuse discarded specimen.
4. Disconnect syringe and discard into sharps container.
5. Scrub the hub with an alcohol pad for 15 seconds and allow to dry.
6. Attach the vacutainer to the needleless connector.
7. Fill vacutainer tubes following the correct sequence for ordered blood tests.
 - a. Allow each tube to fill to the needed volume.
 - b. Gently invert tubes with additives
8. After all tubes are filled and withdrawn from tube holder, detach the holder and discard.
9. Scrub the hub for 15 seconds using friction. Allow to dry.
10. Flush with 20 ml NS, using push-pause method.
11. Disconnect flush and clamp line.
12. If IV fluids were previously infusing, open clamp and resume infusion.
13. Remove gloves and perform hand hygiene.

14. Label specimen per policy.
15. Ensure second witness RN is documented in the electronic health record Hospital at Home will not require second witness.

NOTE: There should be no blood visible in the needleless connector after flushing. The needleless connector may need to be changed if blood cannot be completely cleared.

Appendix F

Adult Blood Sampling from a CVC, Syringe Method

No blood draws from a central line (unless ordered by physician or provider); use blood in lab whenever possible or future draw. Refer to Blood Collection policy for details on order of draws.

Pre-Procedure Preparation

1. Perform hand hygiene before direct contact with patient and subsequently as required throughout procedure steps.
2. Verify patient with two identifiers.
3. Gather supplies, disinfect work area (as applicable).
4. Explain procedure to patient.
5. Ensure second RN is present for a witness. Hospital at Home will not require second RN witness.
6. Stop all infusions through the catheter, clamp lumens, and/or stop infusions as appropriate.

Procedure

1. Scrub the hub with an alcohol pad for 15 seconds prior to each entry of the line and allow to dry.
2. Flush lumen with 0.9% 10 ml sodium chloride before obtaining blood sample using push pause method.
3. Withdraw discard of 5 ml into syringe. If collecting coagulation labs through a catheter with heparin, withdraw discard of 10 ml. Do not reinfuse discarded specimen.
4. Disconnect syringe and discard into sharps container.
5. Scrub the hub with an alcohol pad for 15 seconds and allow to dry.
6. Attach empty 10ml syringe to needleless connector and aspirate the required blood volume.
7. Detach filled syringe.
8. Fill vacutainer tubes following the correct sequence for ordered blood tests. Use a blunt tip needle.
 - a. Allow each tube to fill to the needed volume.
 - b. Gently invert tubes with additives
9. Scrub the hub for 15 seconds using friction. Allow to dry.
10. Flush with 20 ml NS, using push-pause method.
11. Disconnect flush and clamp line.

12. If IV fluids were previously infusing, open clamp and resume infusion.
13. Remove gloves and perform hand hygiene.
14. Label specimen per policy.
15. Ensure second witness RN is documented in the electronic health record. Hospital at Home will not require second RN witness.

NOTE: There should be no blood visible in the needleless connector after flushing. The needleless connector may need to be changed if blood cannot be completely cleared.

Appendix G

Adult Blood Sampling from a CVC In-Line Blood Sampling System

No blood draws from a central line (unless ordered by physician or provider); use blood in lab whenever possible or future draw. Refer to Blood Collection policy for details on order of draws.

For those areas that use the in-line blood sampling system, see procedure below:

1. Pull back on in-line syringe until 5ml of blood has been obtained.
2. Turn the stopcock to the off position.
3. Remove cap vigorously scrub the sample port with a 70% alcohol prep pad for 15 seconds, use friction.
4. Insert vacutainer blunt cannula into sampling port and aspirate the required blood sample.
5. Turn stopcock back to the ON position.
6. Re-infuse patient's blood slowly, no faster than 1ml per second, by pressing the plunger of the in-line syringe back to the closed and locked position.
7. Scrub the hub of the port the specimen was drawn from with alcohol for 15 seconds using friction.
8. Activate the flush device until the line is clear of all blood.